UFSM
Universidade Federal de Santa Maria

PRÁTICA
Simulação de Circuitos com o LTspice®

UFSM00060 – Circuitos Elétricos II
Curso de Engenharia Elétrica
Centro de Tecnologia

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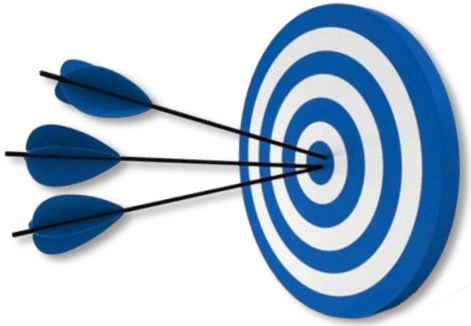
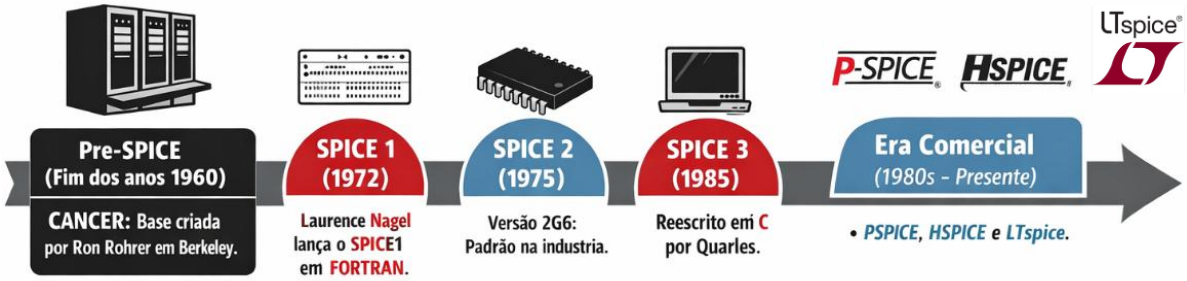


SPICE

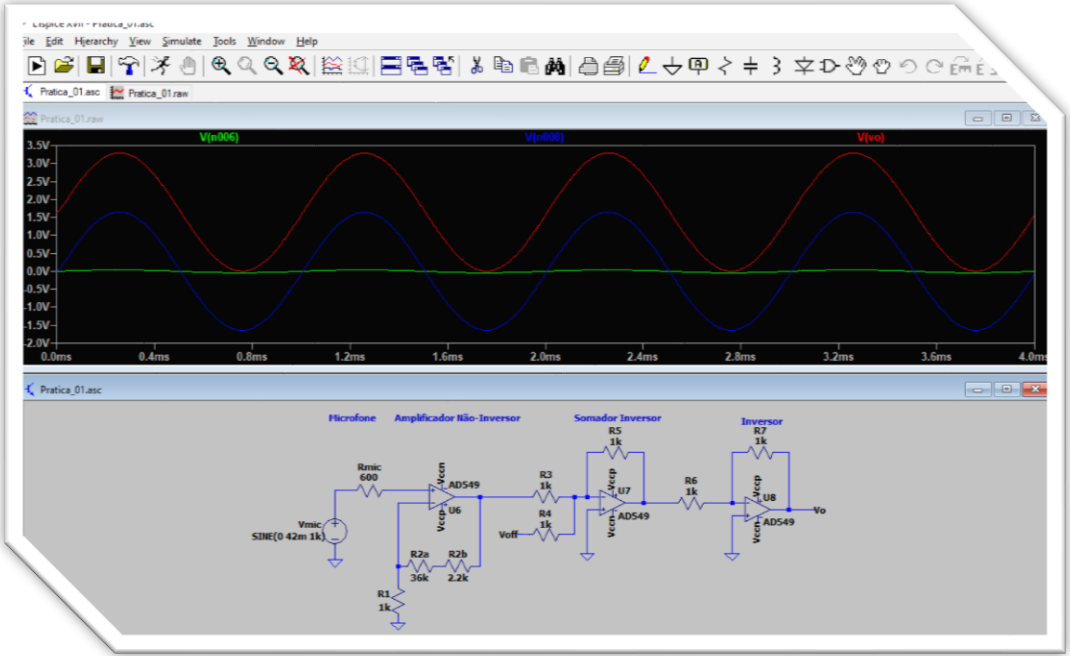
Simulador de circuitos elétricos que utiliza métodos numéricos computacionais para resolver as equações que descrevem o comportamento de circuitos analógicos e digitais



CANCER – *Computer Analysis of Nonlinear Circuits, Excluding Radiation*
SPICE – *Simulation Program with Integrated Circuit Emphasis*

LTspice®

- **Introdução**



LTspice®

- **Características**
  - **Simulador de circuitos elétricos/eletrônicos gratuito**
  - **Permite a visualização de formas de onda**
  - **Permite o emprego de "modelos reais" dos componentes**
    - **Fontes, resistores, capacitores, indutores, transistores, amplificadores operacionais, etc.**

Download

https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html

Manual do usuário

https://www.analog.com/media/en/simulation-models/spice-models/ltspicegettingstartedguide.pdf

LTspice®

- **Atalhos**

Place	
	R – Resistor
	C – Capacitor
	L – Inductor
	D – Diode
	G – GND
	S – Spice Directive
	T – Text
	F2 – Component
	F4 – Label Net
	Ctrl+E – Mirror
	Ctrl+R – Rotate

View	
	Ctrl+Z – Zoom Area
	Ctrl+B – Zoom Back
	Space – Zoom Fit
	Ctrl+G – Toggle Grid
	U – Mark Unncon. Pins
	A – Mark Text Anchors
	Atl+Click - Power
	Ctrl+Click - Attr. Edit
	Ctrl+H – Halt Simulation

Modes	
	ESC - Exit Mode
	F3 – Draw Wire
	F5 – Delete
	F6 – Duplicate
	F7 – Move
	F8 – Drag
	F9 – Undo
	Shift+F9 – Redo

LTspice®

- **Diretivas**

Command	Short Description
.AC	Perform a Small Signal AC Analysis
.BACKANNO	Annotate the Subcircuit Pin Names on Port currents
.DC	Perform a DC Source Sweep Analysis
.END	End of Netlist
.ENDS	End of Subcircuit Definition
.FOUR	Compute a Fourier Component
.FUNC	User Defined Functions
.FERRET	Download a File Given the URL
.GLOBAL	Declare Global Nodes

LTspice®

Command	Short Description
.IC	Set Initial Conditions
.INCLUDE	Include another File
.LIB	Include a Library
.LOADBIAS	Load a Previously Solved DC Solution
.MEASURE	Evaluate User-Defined Electrical Quantities
.MODEL	Define a SPICE Model
.NET	Compute Network Parameters in a .AC Analysis
.NODESET	Supply Hints for Initial DC Solution
.NOISE	Perform a Noise Analysis
.OP	Find the DC Operating Point
.OPTIONS	Set Simulator Options

LTspice®

- **Sufixos e constantes**

Suffix		Suffix	
		f	1e-15
T	1e12	p	1e-12
G	1e9	n	1e-9
Meg	1e6	u	1e-6
K	1e3	M	1e-3
		Mil	25.4e-6

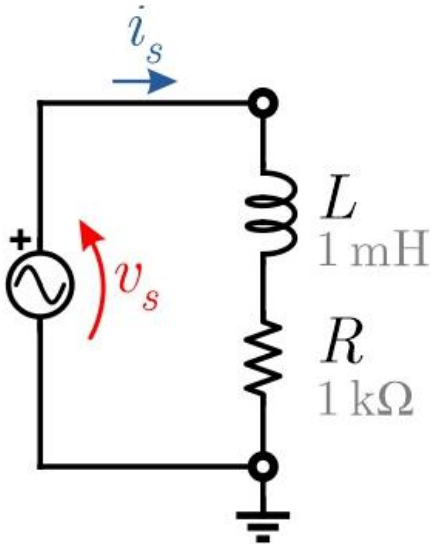
Constants	
E	2.7182818284590452354
Pi	3.14159265358979323846
K	1.3806503e-23
Q	1.602176462e-19
TRUE	1
FALSE	0

Simulação 1 – Circuito RL

- **Objetivos**
  - **Analisar o circuito e determinar o fator de potência**
  - **Realizar a simulação → Analisar a defasagem entre $v_s(t)$ e $i_s(t)$**

$$v_s(t) = V_m \cos(2\pi f_s t)$$

$$\begin{cases} V_m = 1\,\text{V}\ (2\,\text{V}_{\text{pp}}) \\ f_s = 100\,\text{kHz} \end{cases}$$

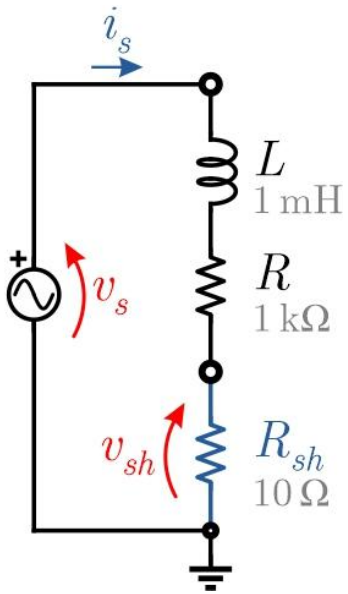


Simulação 1 – Circuito RL

- **Medição de corrente com o osciloscópio (no laboratório)**
  - **Emprego ponteira de tensão e resistor shunt**

$$v_s(t) = V_m \cos(2\pi f_s t)$$

$$\begin{cases} V_m = 1\,\text{V}\ (2\,\text{V}_{\text{pp}}) \\ f_s = 100\,\text{kHz} \end{cases}$$



$$i_s = \frac{v_{sh}}{R_{sh}}$$

Simulação 1 – Circuito RL

- **Análise do Circuito**
  - **Fonte de alimentação**

$$\vec{V}_s = 1\angle 0^{\circ}\ \text{V} \qquad \left(\omega_s = 628{,}318\ \text{k rad/s}\right)$$

$$\boxed{\vec{V}_{s,rms} = 0{,}707\angle 0^{\circ}\ \text{V}}$$

  - **Reatância indutiva da carga**

$$X_L = \omega_s L = 628{,}318\ \Omega$$

  - **Impedância da carga**

$$\boxed{\bar{Z} = R + jX_L = 1000 + j628{,}318\ \Omega = 1{,}181\angle 32{,}14^{\circ}\ \text{k}\Omega}$$

Simulação 1 – Circuito RL

- **Análise do Circuito**
  - **Potência Aparente Complexa**

$$\vec{S} = \frac{V_{s,rms}^2}{\bar{Z}^*} = \underbrace{358,478}_{P} + j\underbrace{225,238}_{Q}\ \mu\text{VA}$$

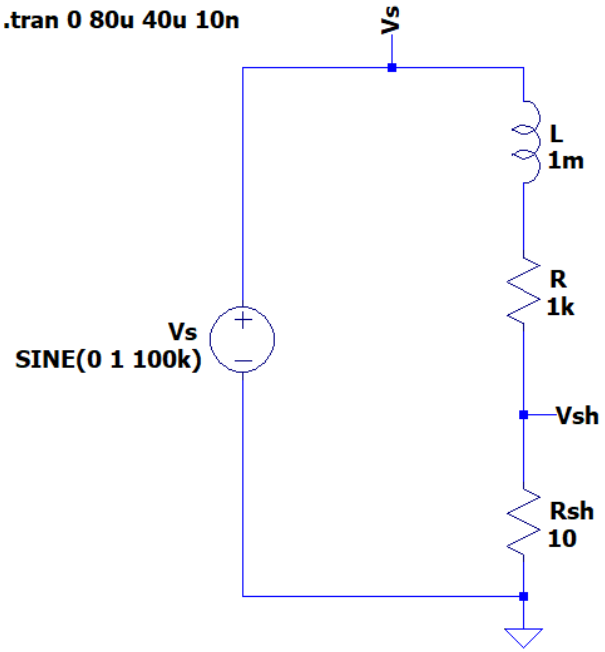
$$S = 423,366\ \mu\text{VA}$$

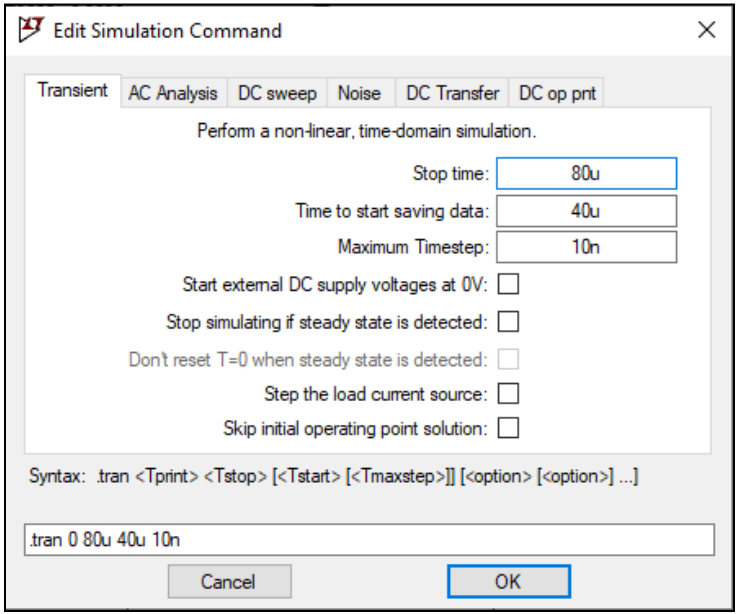
  - **Fator de potência**

$$FP = \frac{P}{S} = \cos\left(32,14^{\circ}\right) \quad \rightarrow \quad \boxed{FP = 0,846}$$

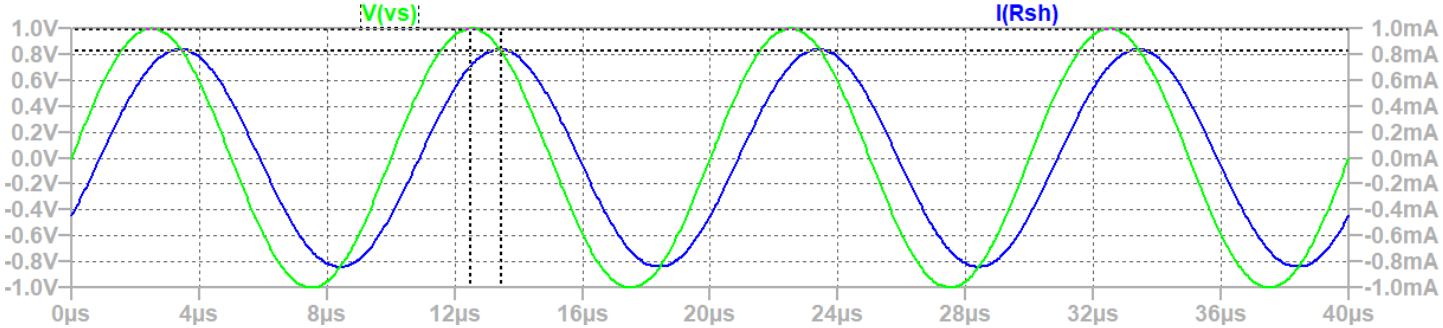
Simulação 1 – Circuito RL

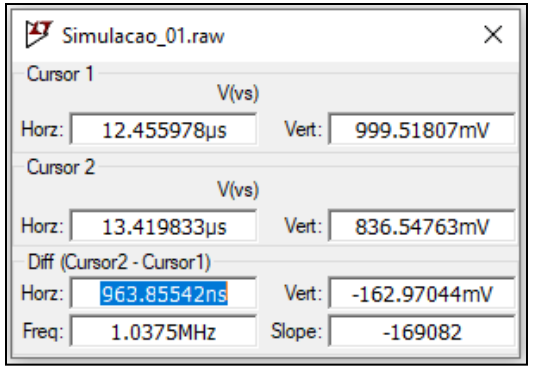
- **Simulação no LTspice**





Simulação 1 – Circuito RL

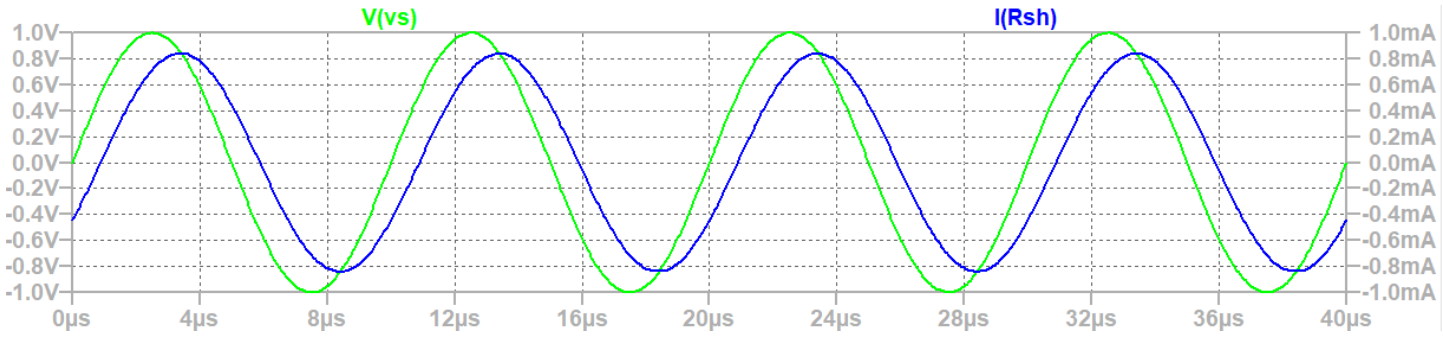


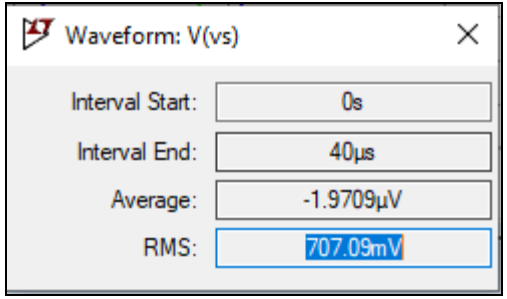


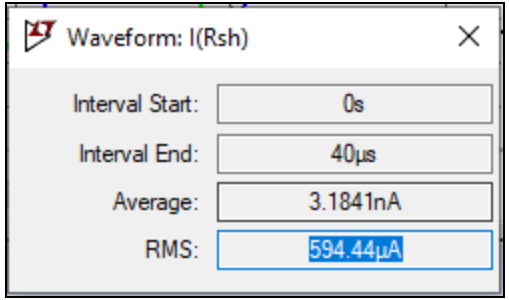
$$\theta = \omega_s \cdot \Delta t = \omega_s \cdot 963{,}85\text{n}$$

$$\theta = 0{,}605 \text{ rad} = 34{,}698^{\text{o}}$$

Simulação 1 – Circuito RL



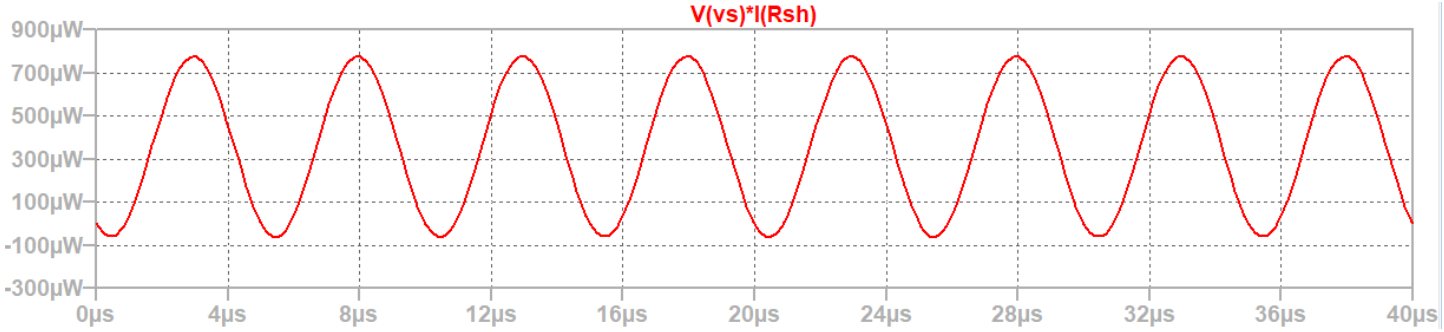


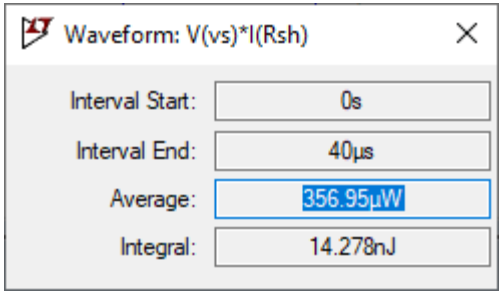


$$S = V_{s,rms} \cdot I_{sh,rms}$$

$$S = 420,286\ \mu\text{VA}$$

Simulação 1 – Circuito RL





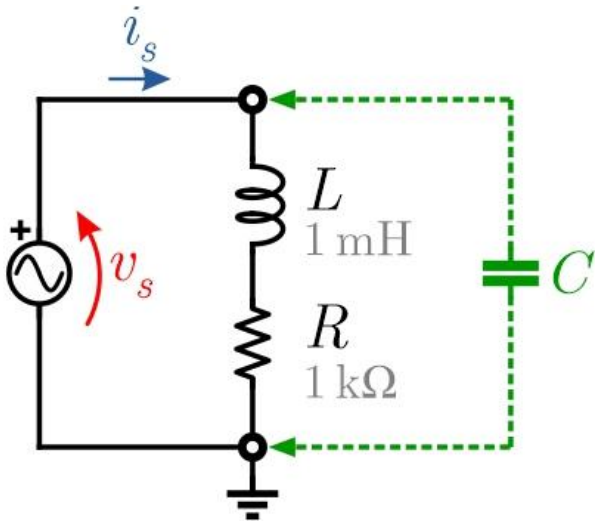
$$P = 356{,}95\ \mu\mathrm{W}$$

Simulação 2 – Correção do Fator de Potência

- **Objetivos**
  - **Projeto do "banco de capacitores" para correção do fator de potência (FP = 1)**
  - **Realizar a simulação → Analisar a defasagem entre $v_s(t)$ e $i_s(t)$**

$$v_s(t) = V_m \cos(2\pi f_s t)$$

$$\begin{cases} V_m = 1\,\text{V}\ (2\,\text{V}_{\text{pp}}) \\ f_s = 100\,\text{kHz} \end{cases}$$



Simulação 2 – Correção do Fator de Potência

- **Medição de corrente com o osciloscópio**
  - **Emprego ponteira de tensão e resistor shunt**

$$v_s(t) = V_m \cos(2\pi f_s t)$$

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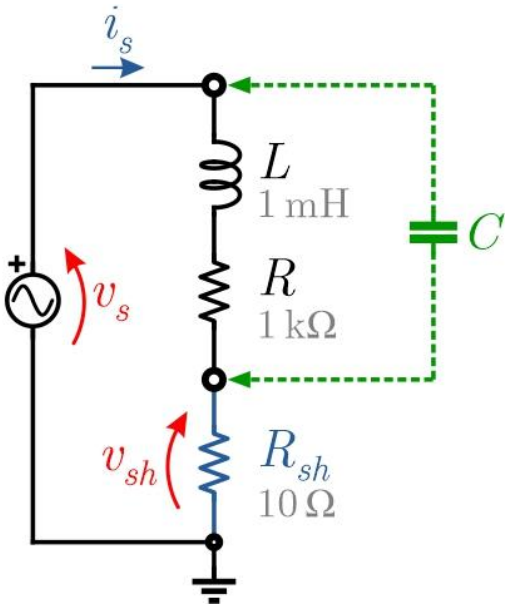


$$i_s = \frac{v_{sh}}{R_{sh}}$$

Simulação 2 – Correção do Fator de Potência

- **Análise do Circuito**
  - **Para obter fator de potência unitário**

$$\boxed{Q_C = Q = 225,238\ \mu\text{VAr}}$$

  - **Como o capacitor está em paralelo com a fonte de alimentação**

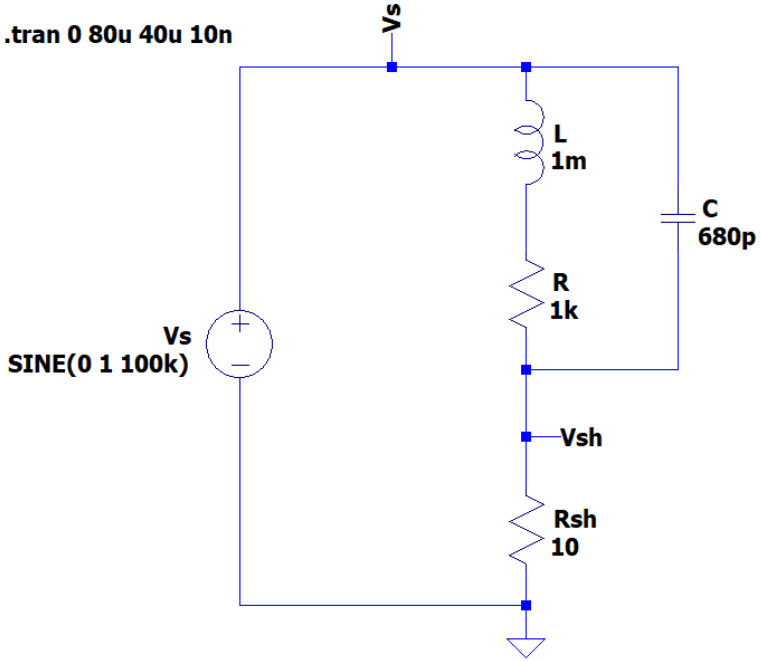
$$Q_C = \frac{V_{s,rms}^2}{X_C} \quad \rightarrow \quad X_C = 2,219\ \text{k}\Omega$$

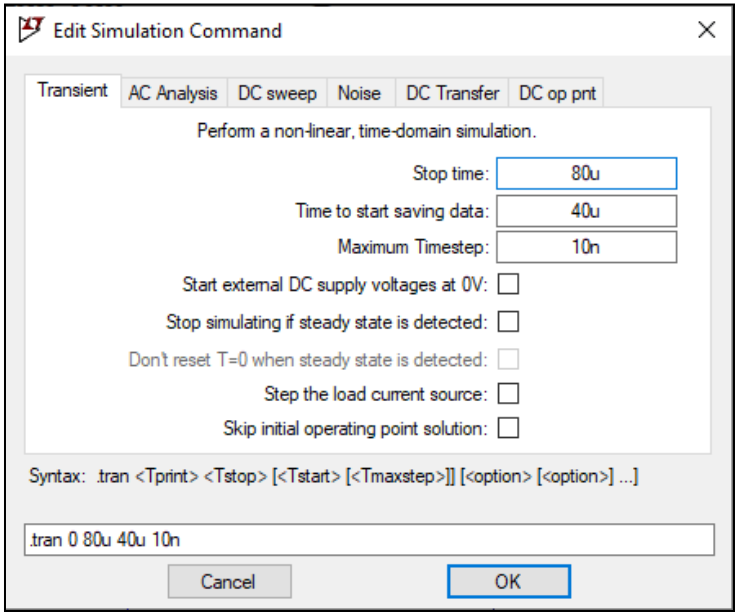
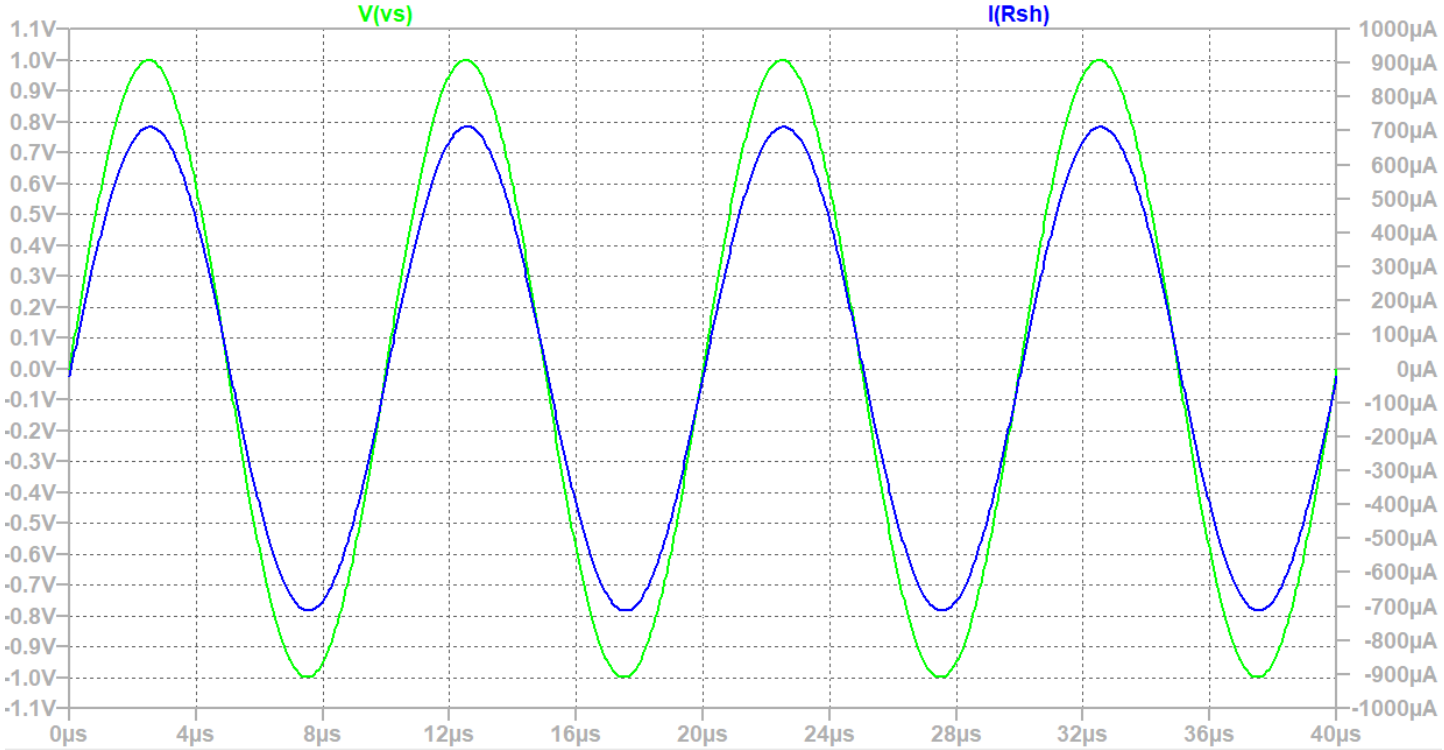
  - **Capacitância**

$$X_C = \frac{1}{\omega_s C} \quad \rightarrow \quad \boxed{C = 716,956\ \text{pF}}$$

Simulação 2 – Correção do Fator de Potência

- **Simulação no LTspice**

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